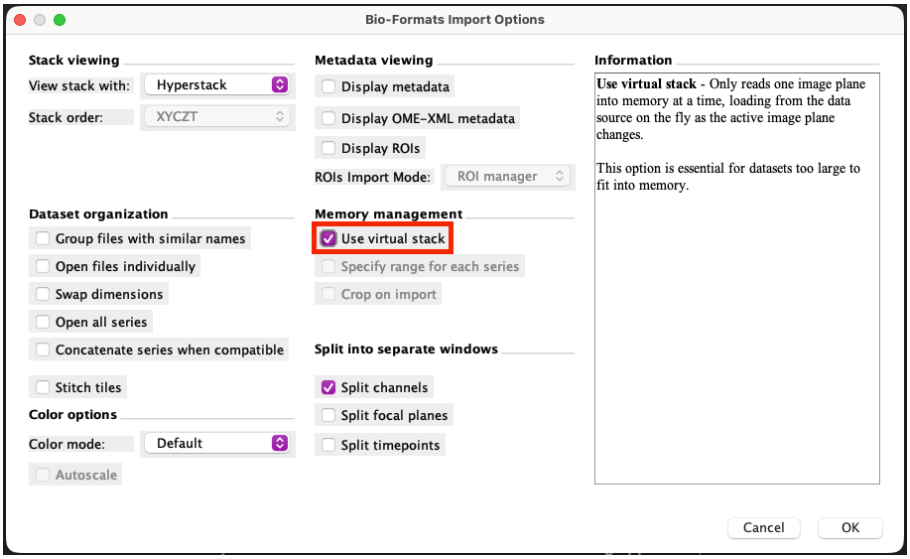
Image Processing Instructions

Last Updated: 10/19/2026 by Clara Shin (clara.shin@wsu.edu)

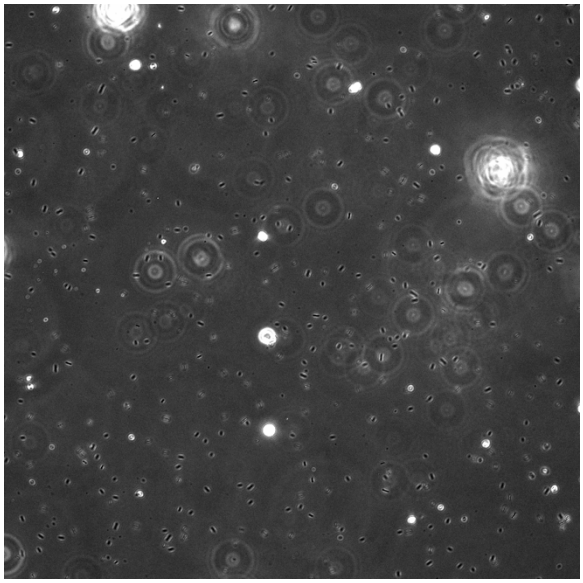
Before running TumbleWeed, the microscopic videos needed to be adjusted, to make the brightness ideal for tracking and having a TIFF format.

1. **Open ImageJ or Fiji and open video file**
    a. File → Open… and browse for a video. In Baylink Lab we save videos as .nd2 files.
    b. Make sure to check "use virtual stack" not to overload your computer's memory.

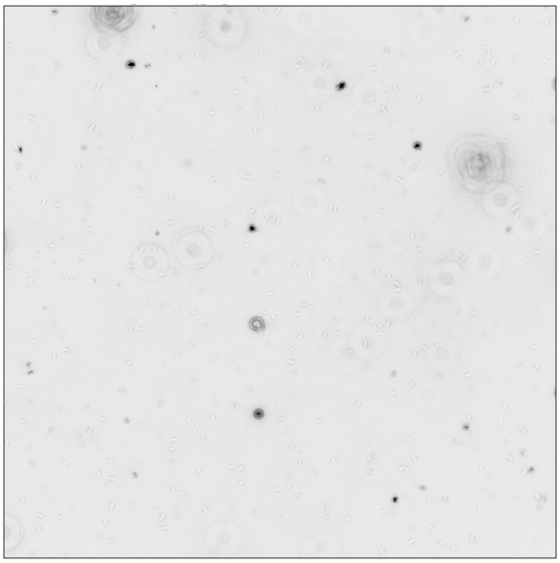


2. **Create Duplicate Stacks**
    a. Image → Adjust → Brightness/Contrast… → Click Auto
    b. Image → Duplicate… → check "Duplicate stack". When finished, a new window will open and have a name on the top ending "C=0-1"

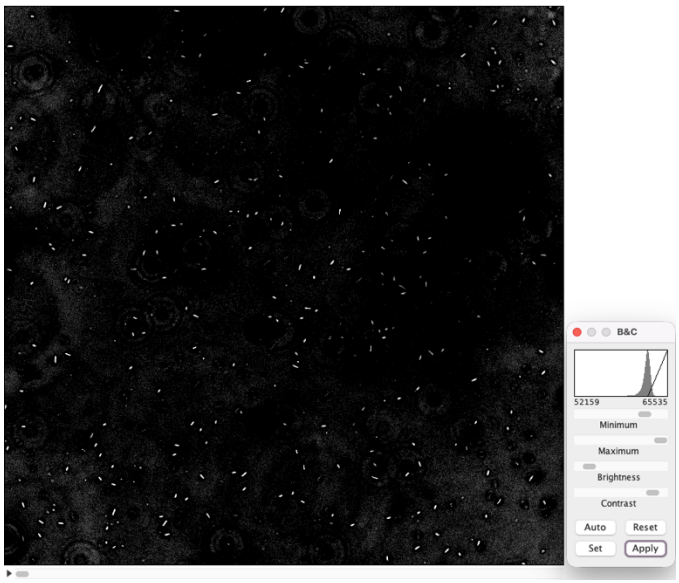




3. **Invert Duplicated Images** (Skip to Step 4 if cells were imaged with fluorescence)
    a. Edit → Invert → Yes. The images should look like this:



4. **Adjust Brightness & Contrast of Duplicate stacks**
    a. Use the Brightness & Contrast sliders to make the cells visible with a dark background. Start with using Auto → Apply then slide the Minimum bar to achieve the image brightness/ contrast you'd like. Click **Apply** (Important!)



5. **Save**
    a. File → Save As → Image Sequence…
    b. Browse a folder (make a dedicated folder for the image sequence)
    c. Format: TIFF

Reference: Swim Tracking Tutorial by Kaily Franco (link)